Computational Statistical Techniques: Applications of MCMC Algorithms in Distribution Simulation Problems

MSC IN APPLIED STATISTICS
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Metropolis Algorithm: Analysis of Acceptance and Rejection Streaks

For simulating from the distribution with density proportional to:

$$\pi(x) \propto \frac{e^{-x^2}\sin(x^2)}{1+x^2}, \qquad x \in R$$

the Metropolis algorithm was applied with a standard normal proposal distribution $\mathcal{N}(0,1)$. The Markov chain was run for $N = 10000$ steps, and the first 1000 were discarded as burn-in.

1. Acceptance rate and effective sample size

The implementation yielded an acceptance rate of **49.05%**. The effective sample size (ESS) was calculated as **1145.7**, indicating that, despite autocorrelation, the chain provides useful information about the target distribution.

2. Estimation of $P(X > 1)$

Based on the post-burn-in samples, the probability that X exceeds 1 is estimated to be **0.1369**.

3. Analysis of rejection streaks (run-lengths)

In the Metropolis algorithm, when a proposal is rejected the chain stays in the same state, potentially creating sequences of consecutive rejections (rejection streaks). To study this phenomenon, the implementation was modified to record, for each step (after burn-in), whether the proposal was rejected. From the resulting logical vector, the lengths of all consecutive rejection sequences (run-lengths) were computed using the *rle* function. The distribution of these lengths is shown in the following table:

Rejection Run-Length Distribution			
Length	Frequency	Length	Frequency
1	1196	8	18
2	630	9	8
3	292	10	5
4	154	11	5
5	77	12	2
6	37	13	1
7	26		

Table 1: Frequency distribution of rejection run-lengths

Nearly half (48.8%) of the rejection streaks have length 1, meaning that rejections are mostly isolated and the chain does not remain stuck for many consecutive steps. The frequencies decrease sharply and consistently, suggesting exponential decay. Only 13.6% of the streaks are longer than 3 steps, and fewer than 1% exceed 10 steps. This indicates that the chain avoids prolonged periods of stagnation and explores the state space effectively. The smooth and regular decrease in frequencies, without unusual peaks or gaps, shows that the standard normal proposal is well-matched to the target distribution. The acceptance rate around 50%, together with the run-length distribution,

confirms that the proposal is neither too conservative (which would lead to high acceptance but low mobility) nor too aggressive (which would cause many rejections and long stagnation sequences).

Metropolis with Uniform Proposal: Simulation and Convergence

For simulating from the distribution with density proportional to:

$$\pi(x) \propto sin^4(x+3)cos^2(x-4)exp\left(-\frac{(x-1)^2}{2}\right), \qquad x \in R$$

the Metropolis algorithm was applied with a uniform proposal distribution $U(-a, a)$. After experimentation, the parameter was set to $a = 3.1$, so that the proposal width allows satisfactory chain mobility. The Markov chain was run for $N = 100000$ steps, with the first 10000 discarded as burn-in.

1. Simulation and convergence assessment

The implementation followed the standard Metropolis procedure:

- Initial value $x_1 = 2.5$.
- At each step m, a candidate $y = x_{m-1} + z$, where $z \sim U(-3.1,3.1)$.
- Acceptance probability $\alpha = \min(1, \frac{\pi(y)}{\pi(x_m - 1)})$

After discarding the first 10000 values, the remaining 100000 samples were used for analysis.

2. Histogram and theoretical density

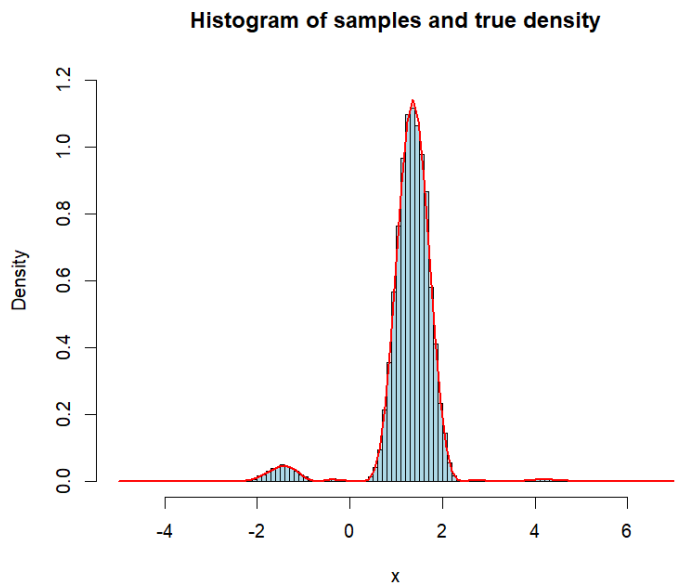

Figure 1: Histogram of Metropolis samples (blue) and theoretical density π(x) (red curve)

To visualise the convergence, the normalising constant of $\pi(x)$ was computed numerically by integration, and a histogram of the samples was constructed together with the normalised theoretical density. The histogram shows that the empirical distribution of the samples closely matches the theoretical curve, confirming that the chosen proposal distribution $U(-3.1, 3.1)$ and the Metropolis algorithm produce samples that follow the desired target distribution.

3. Convergence of the sample mean and variance

For studying convergence, the following quantities were computed the true mean μ and true variance σ^2 of the distribution, via numerical integration. The cumulative sample means and cumulative sample variances.

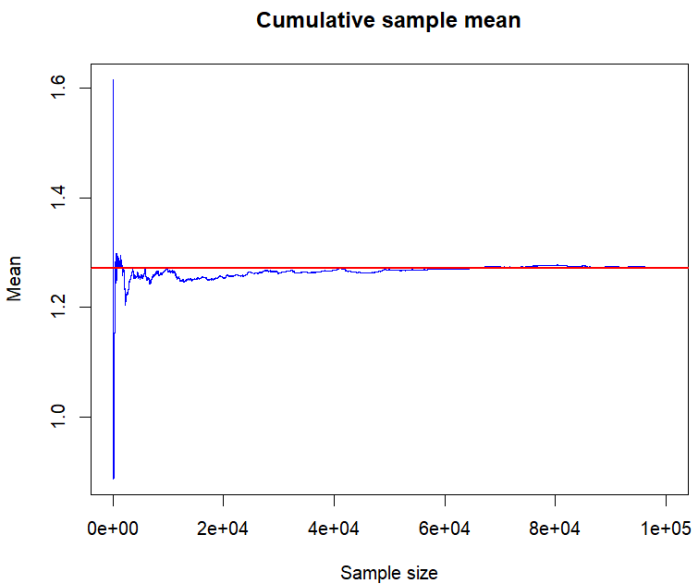

Figure 3: Convergence of the cumulative sample mean (blue line) to the true mean (red line)

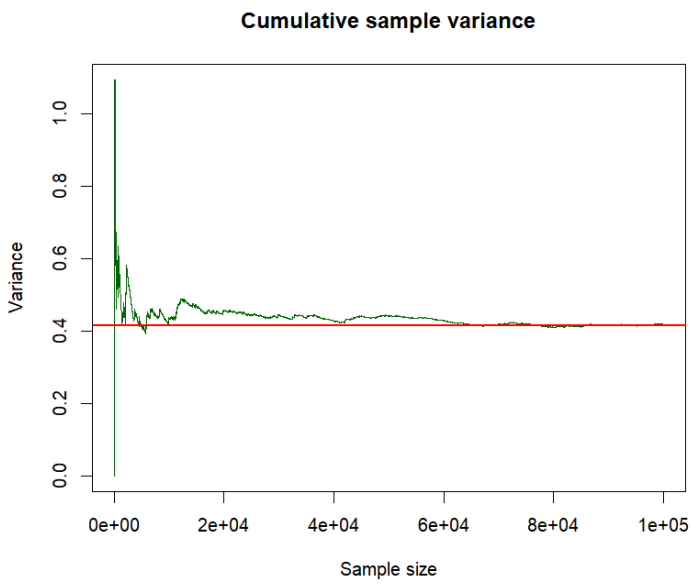

Figure 2: Convergence of the cumulative sample variance (green line) to the true variance (red line)

In the two presented plots, the blue line shows the cumulative estimate and the red line shows the true value. In both cases the estimate converges quickly to the true value, confirming the correct behaviour of the algorithm.

The choice $a = 3.1$ resulted in good exploration of the state space, as evidenced by the close overlap between the histogram and the theoretical density. The convergence of the sample mean and variance was smooth and fast, without long transient periods. Overall, the Metropolis algorithm with a uniform proposal proved effective for this distribution, producing a representative sample after burn-in.

Metropolis-Hastings with Logarithmic Transformation for a Positive Variable

For simulating from the distribution with density proportional to:

$$\pi(x) \propto \left(\frac{x^2}{2} + 3x\right) e^{-5x}, \qquad x > 0$$

the Metropolis-Hastings algorithm was applied with a normal proposal distribution. Due to the positive support of X, a random walk was applied to the transformed variable $Y = \log X$. The proposal is made in the Y space as $Y' \sim \mathcal{N}(Y, \sigma^2)$, with $\sigma = 1$, and acceptance is computed using the ratio of the densities with the appropriate Jacobian adjustment. The chain was run for $N = 100000$ steps, with the first 1000 discarded as burn-in.

1. Histogram and theoretical density

The normalising constant of $\pi(x)$ was computed numerically, and a histogram of the samples was constructed together with the normalised theoretical curve. The empirical distribution of the samples closely matches the theoretical density, confirming the correct behaviour of the algorithm.

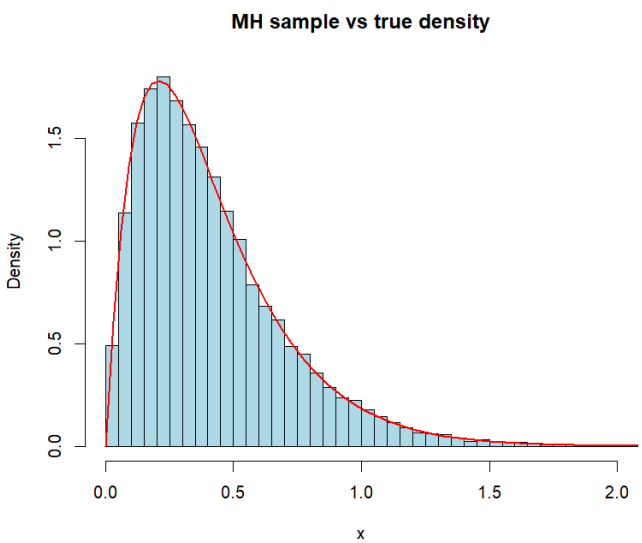

Figure 4: Histogram of Metropolis-Hastings samples (blue) and theoretical density π(x) (red curve)

2. Estimation of mean and median

From the post-burn-in samples, the estimates obtained were:

- Mean $\hat{\mu} \approx 0.396$
- Median $\widehat{m} \approx 0.324$

The convergence plot shows the cumulative estimates of the mean and median for $n = 1, \dots, N$. Both curves stabilise quickly, indicating that the chain converges effectively.

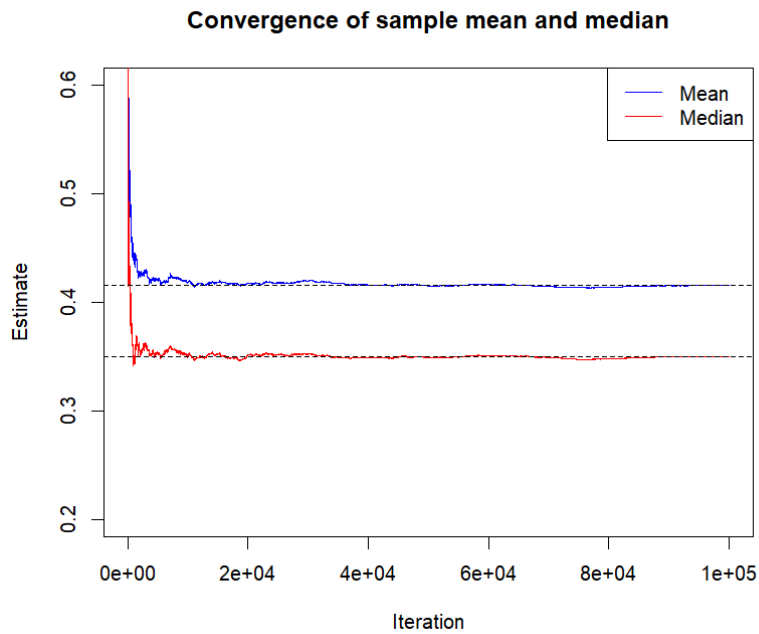

Figure 5: Convergence of the sample mean (blue) and median (red) as the sample size increases

Metropolis-Hastings with Normal Proposal for a Distribution with Polynomial Exponent

For simulating from the distribution with density proportional to:

$$\pi(x) \propto \exp\left\{-0.5\left[\left(\frac{x}{10}\right)^4 + \left(\frac{x}{10}\right)^2 + x^2\right]\right\}, \qquad x \in R$$

the Metropolis-Hastings algorithm was applied with a normal proposal distribution $\mathcal{N}(x, \sigma^2)$. After experimentation, $\sigma = 2.7$ was chosen to achieve a satisfactory acceptance rate. The Markov chain was run for $N = 10000$ steps, with the first 1000 discarded as burn-in.

1. Results

The implementation yielded:

- Sample mean: $\hat{\mu} \approx -0.007$

- Acceptance rate: 40.1%
- Effective sample size (ESS): 1811

2. Histogram and theoretical density

The normalising constant of $\pi(x)$ was computed numerically, and a histogram of the post-burn-in samples was constructed together with the normalised theoretical curve. The title of the figure displays the sample mean, acceptance rate, and effective sample size.

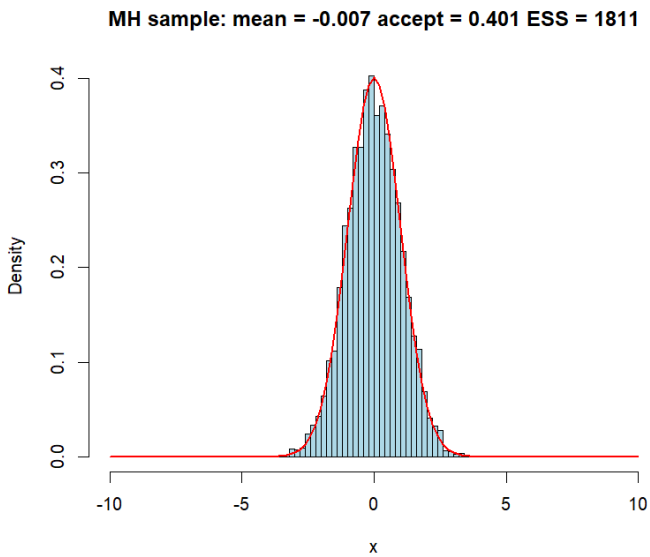

Figure 6: Histogram of Metropolis-Hastings samples (blue) and theoretical density (red curve). The title displays the sample mean, acceptance rate, and effective sample size.

The empirical distribution of the samples closely matches the theoretical density, confirming the correct behaviour of the algorithm. The acceptance rate around 40% and the relatively high ESS indicate that the chain explores the space effectively, despite autocorrelation.

Gibbs Sampler for a Bivariate Exponential Distribution

For simulating from the joint distribution:

$$\pi(x,y) \propto e^{-5x(1+y)-y(3+2x)}, \qquad x,y > 0$$

the Gibbs sampler was applied, since the conditional distributions are known and easy to sample from:

$$\pi(x|y) \propto e^{-5x-7xy} = e^{-x(5+7y)}, \qquad x|y \sim Exp(5+7y)$$

$$\pi(x|y) \propto e^{-3y-7xy} = e^{-y(3+7x)}, \qquad y|x \sim Exp(3+7x)$$

The chain was run for $N = 100000$ iterations, with the first 10000 discarded as burn-in. After applying the algorithm, the distributions of X, Y and $Z = X + Y$ are shown below.

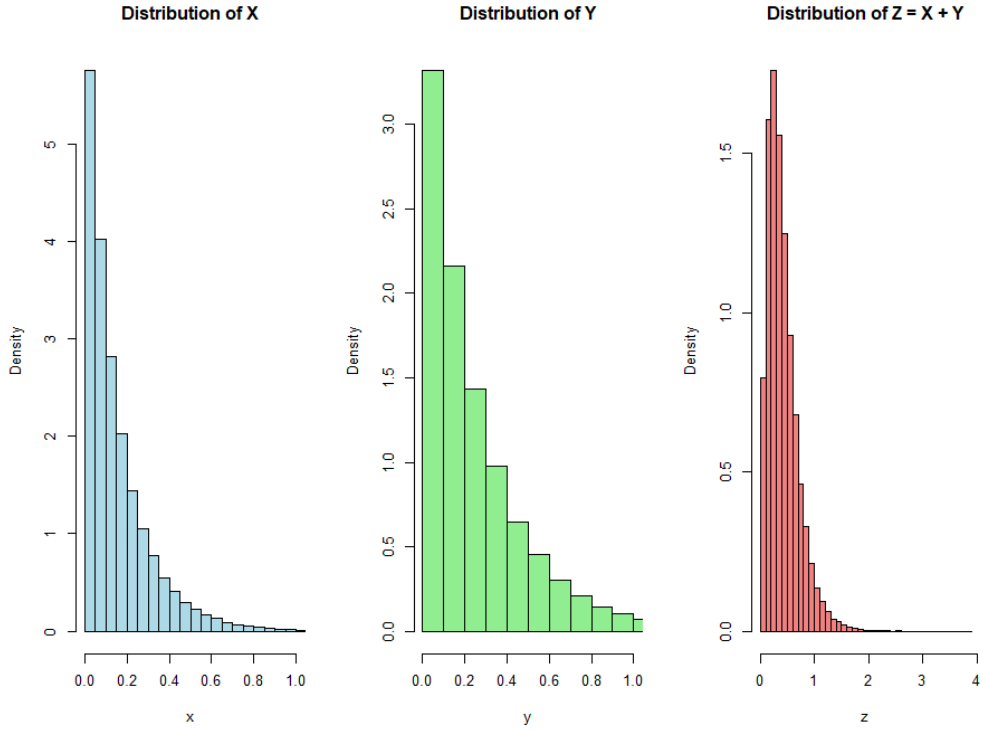

Figure 7: Histograms of the Gibbs sampler samples for the marginal distributions of X, Y and the total quantity Z=X+Y

From the simulation of $N = 100000$ samples (with burn-in 10000), the estimated mean and variance of the total quantity Z are:

- Mean $\hat{\mu}_Z \approx 0.4143$
- Variance $\hat{\sigma}_Z^2 \approx 0.0827$

Their convergence is shown below.

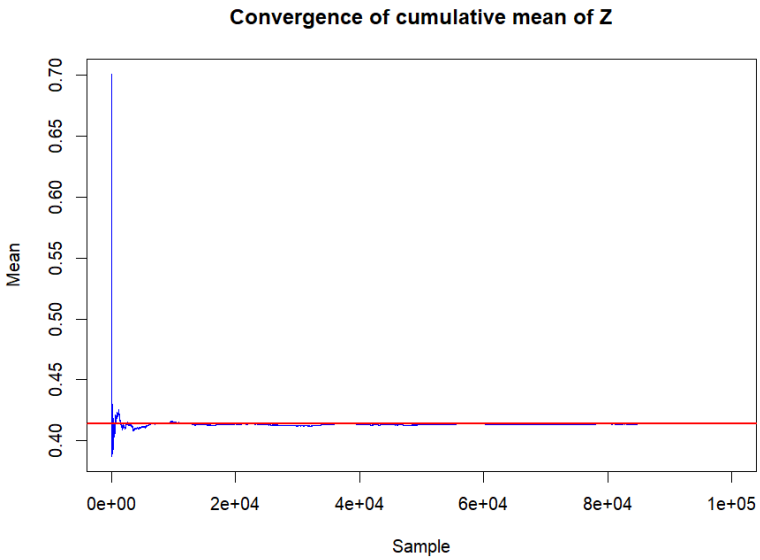

Figure 8: Convergence of the sample mean of Z

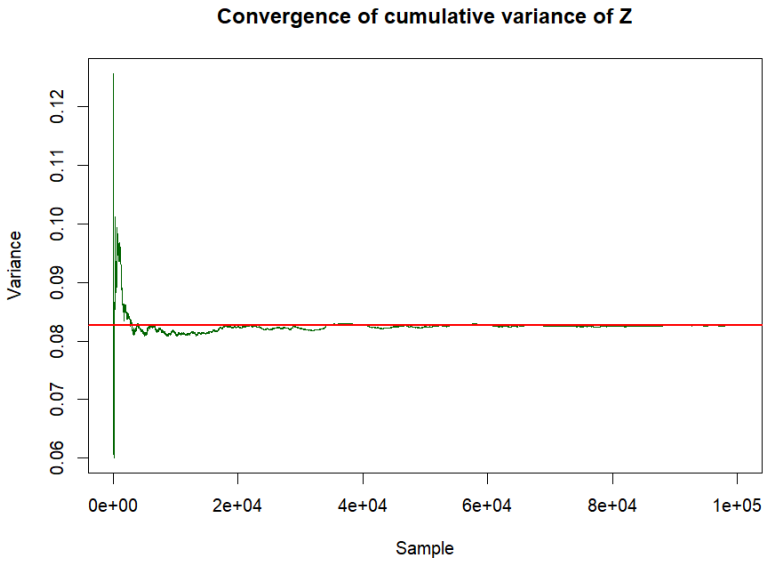

Figure 9: Convergence of the sample variance of Z

Gibbs Sampler for a Distribution with Normal and Gamma Conditionals

For simulating from the joint distribution:

$$\pi(x,y) \propto \sqrt{y} \times exp\left\{-\frac{y}{2}[1+(x-\pi)^2]\right\}, \qquad x \in R, y > 0$$

the Gibbs sampler was applied with conditional distributions:

$$x|y \sim N\left(\pi, \frac{1}{y}\right)$$

$$y|x \sim G\left(\frac{3}{2}, \frac{1+(x-\pi)^2}{2}\right)$$

The chain was run for $N = 100000$ iterations, with the first 10000 discarded as burn-in.

1. Estimation of $E\left[\frac{X}{1+Y}\right]$

From the post-burn-in samples, cumulative estimates of the quantity [X/1+Y] were computed. The presented plot shows the convergence of the estimate as the sample size increases.

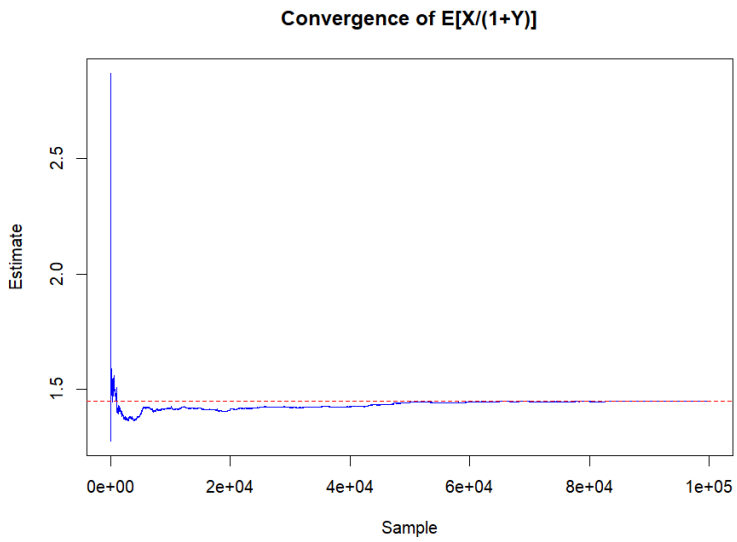

Figure 10: Convergence of the estimate of E[X/(1+Y)] as the sample size increases.

2. Non-existence of $E\left[\frac{X}{Y}\right]$

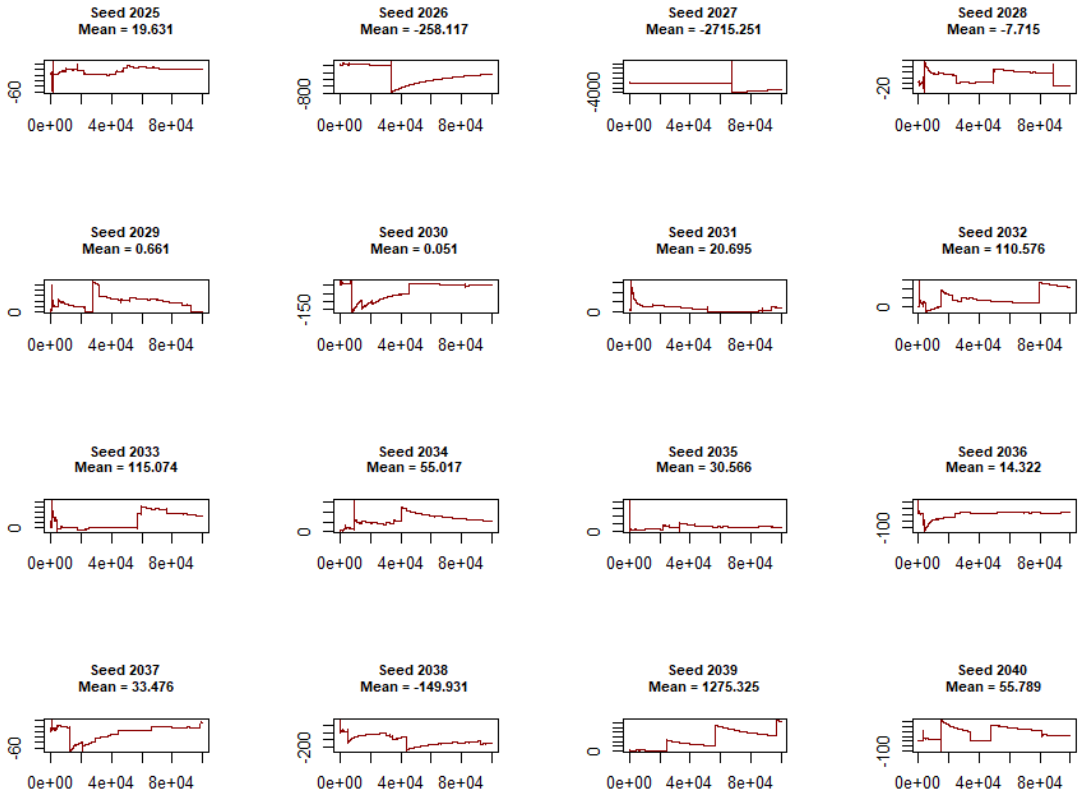

Figure 11: Cumulative estimates of E[X/Y] for 16 different random seeds (2025–2040). The lack of convergence confirms the non-existence of the mean

The mean E[X/Y] does not exist (is not finite) for the following reason:

When Y takes very small values (approaches zero), the quantity $|X/Y|$ becomes extremely large. From mathematical analysis, the marginal distribution of Y is such that the integral $\int(1/y)\ \pi(y)\ dy$ diverges to infinity. Therefore, the mean of the absolute quantity $E[|X/Y|]$ is infinite, and consequently $E[X/Y]$ is not defined as a finite number. To demonstrate this instability experimentally, 16 independent simulations were performed with different random seeds (2025 to 2040). For each, the cumulative estimate of $E[X/Y]$ was computed. The 16 plots show that the estimate does not converge but exhibits large, unpredictable fluctuations, confirming that the mean does not exist.